

# AI Winters: Boom and Bust — presenter guide

One HTML file, offline, no AI at runtime. You need: a screen to project or share; no other materials.

## Learning goals

Not stated objectives — what the activity itself demonstrably shows, one per stage:

- 1. A forecast needs an outcome and a deadline; an assessment needs evidence about its own moment — grading one as the other makes history look more certain than it is.
- 2. The five lanes organize evidence for comparison; they do not prove that every AI contraction follows one fixed order.
- 3. An open cell is evidence only that a limit or a withdrawal has not (yet) been established — not a prediction that one is coming.
- 4. A historical rhyme is a reason to ask a question, not a reason to expect the same ending — each claim about today needs its own evidence.

## Where this sits on the path

Demo 3 leaves the room reading a benchmark result under its own stated conditions rather than extrapolating from two points. Fast progress like that is exactly what raises the question this demo answers: *is this a bubble?* History answers with evidence, not a script — and this demo hands forward the habit of classifying a claim, forecast or assessment, before judging it.

## One question for the session

**Has this happened before?** AI has contracted twice, through different combinations of technical limits, institutional choices and markets. The histories rhyme; neither is a script for the present.

*The one sentence to land: **a prediction needs a deadline and an outcome; an assessment needs evidence about its own moment.** Do not grade a historical claim until you know which kind it is.*

## The shape of the session

| STAGE       | WHAT HAPPENS                                                                                                                                                           | TIME    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 — Claims  | Five time-bounded forecasts, four assessments or untimed ambitions, and one warning, with the speaker and date hidden. Score forecasts; classify the unscored context. | 7–9 min |
| 2 — History | Scrub the timeline. Five lenses compare historical pathways without forcing a sequence.                                                                                | 3–4 min |
| 3 — Compare | Compare the two winters and the present under the same lenses. Two current questions stay open.                                                                        | 2–3 min |
| 4 — Today   | Rhymes against differences. Close on the checklist.                                                                                                                    | 2–3 min |

## Stage 1 — Claims

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**Question.** Is this a forecast you can grade, or an assessment you can only judge by its own moment?

**Predict prompt.** Guess the year, then say whether the claim is a scoreable forecast or an assessment to classify — before Lock it in reveals anything.

**Activity.** Do the first two cards yourself, out loud, so the room sees the mechanic. Then hand it over — ask for a show of hands on the year before you drag the slider, and a verbal verdict or classification before you press **Lock it in**. For the four context cards, choose **Assessment or untimed ambition** when the claim supplies no scoreable future window. The room being wrong together is the experience; a silent room reading a screen is not.

**Expected observation.** Three of the five time-bounded promises missed outright, one arrived thirty years late, and the deliberately unscored ones (the 1958 press claim, ALPAC, Perceptrons, Lighthill, and Amodei 2024) pull the room toward wanting to grade them anyway — that pull *is* the misconception to name.

**Check-yourself answer key.** "Should an open forecast be marked wrong?" — No: an unresolved forecast stays unscored rather than counted as a failure.

**Teaching limit.** This is a curated ten-claim deck, not a representative accuracy sample of optimists, pessimists, or the field.

## The deck, and what each card is for

| YEAR | CARD                                                | TYPE             | WHAT IT IS FOR                                                                                                                               |
|------|-----------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1950 | Turing — fifty years to the imitation game          | OPEN FORECAST    | Its about-2000 bar was missed; the current status remains contested. Use it to separate a past deadline from an unresolved present judgment. |
| 1958 | “walk, talk, see, write, reproduce itself”          | UNTIMED AMBITION | An expansive press claim without a deadline. Classify it; do not pretend it supplies a scoreable outcome window.                             |
| 1957 | Simon & Newell — chess champion in ten years        | PROMISE          | <b>The most useful card in the deck.</b> Right destination, wrong by a factor of four. Pause here.                                           |
| 1960 | Simon — any work a man can do, in twenty years      | PROMISE          | Twenty years lands in 1980, the bottom of the first winter. Also: it is usually misdated to 1965.                                            |
| 1966 | ALPAC — machine translation is not near             | ASSESSMENT       | Distinguish a near-term economic assessment from a forecast. The report also recommended continued computational-linguistics research.       |
| 1967 | Minsky — substantially solved within a generation   | PROMISE          | A reported quotation attributed to his textbook. Use it to explain why the app labels this source route reported.                            |
| 1969 | <i>Perceptrons</i> — neural nets read as a dead end | ASSESSMENT       | The proofs were right; the history of how the field received them is disputed.                                                               |
| 1973 | Lighthill — no major impact so far                  | ASSESSMENT       | A retrospective judgment on impact and constraints, not a permanent forecast.                                                                |
| 1984 | Schank & Minsky — a winter is coming                | WARNING          | The card that saves the demo from being anti-hype. They were <b>right</b> , three years early.                                               |
| 2024 | Amodei — “could come as early as 2026”              | OPEN             | Deliberately unscored. The year named is the year you are standing in.                                                                       |

**Short of time?** Keep four: **Simon & Newell 1957** (over-promise, right destination), **ALPAC 1966** (assessment versus forecast), **Schank & Minsky 1984** (a prediction that was right), and **Amodei 2024** (still open). Skip to Stage 3 and close.

## Stage 2 — History

**Question.** Do the two AI winters share one fixed cause-and-effect sequence, or different pathways that happen to rhyme?

**Predict prompt.** Scrubbing from 1950, does an institutional bet or a named constraint fire first in each winter?

**Activity.** Drag the slider slowly, or step with ‘ and ’ to stop and read one event. Tap any dot for its full text and source. Use the rows as comparison lenses — claims, institutional bets, constraints, reassessment, withdrawal — and ask which causes matter in each episode and which do not carry across. Read the current entries as evidence about today, not a third historical run.

**Expected observation.** The order is not identical between the two winters: the first winter's reassessment (ALPAC, then Lighthill) came from commissioned reports years apart, while the second winter's reassessment was the market itself — the LISP hardware collapse, with no equivalent report. Two histories, not one template.

**Check-yourself answer key.** "Does a named bet before a withdrawal in both winters prove a rule?" — No: two episodes sharing an order is not a proven causal sequence.

**Teaching limit.** No continuous, sourceable AI funding series exists across these years; four discrete funding examples with different currencies and accounting bases are shown instead of one invented total.

*Do not let the lenses turn into a prophecy. The correct line is: "these comparisons tell us what to investigate. They do not establish that a withdrawal will occur."*

## Stage 3 — Compare

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**Question.** Which of the five comparison lenses still shows a real similarity once you separate description from prediction?

**Predict prompt.** On the "Constraints" lens, is "Today" an established limit, or an open question?

**Activity.** Choose one of the five lens buttons; its navy fill marks the selection. Compare the three era headings, then open each era's **Details** for the full explanation and named sources. Escape closes the panel and returns focus to its button. Compare causes, not step numbers.

**Expected observation.** Two cells in the "Today" column stay open — no single binding technical constraint is established, and no comparable broad contraction is established either — while the claims, money and reassessment lenses do resolve to sourced present-day evidence.

**Check-yourself answer key.** "Does an open cell mean a winter is currently impossible?" — No: open means not yet established either way, not ruled out.

**Teaching limit.** Calling a present-day cell "open" is this page's own reading of the evidence available at build time, not a citation that says "unresolved."

*"Hindsight makes old constraints easier to see. It does not prove that today has the same constraint or that it must end the same way. Ask what evidence would support the claim."*

## Stage 4 — Today

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**Question.** Which rhyme still holds once you weigh it against a genuine difference?

**Predict prompt.** Pick one rhyme and one difference — does the rhyme survive once you weigh the difference against it?

**Activity.** Expand a similarity and a difference, giving both columns roughly equal time; the supporting statements start closed so the room investigates one claim at a time. Close on the checklist, not a verdict: is it a forecast or an assessment, what evidence supports it, and what evidence would move it.

**Expected observation.** The strongest rhyme (an evaluation warning about SWE-bench Verified) and the strongest difference (measured scaling behaviour and reported consumer reach) do not cancel each other — each still needs its

own evidence, which is the point.

**Check-yourself answer key.** "Does a bigger user base rule out a contraction?" — No: user count answers a different question than whether funding, evaluation trust, or returns hold up.

**Teaching limit.** Historical comparison is not a prediction; this curated set of near-term forecasts that missed does not establish a representative error rate for optimists, pessimists, or AI research generally.

## Misconceptions

### "An unresolved forecast should count as a failure."

No. Only a forecast whose window has closed one way or the other gets a verdict; an open claim stays unscored (Stage 1).

### "The same order appearing in both winters proves a fixed sequence."

No. Two historical cases sharing a pattern is a reason to look closer, not a general law about every contraction (Stage 2).

### "An open cell in the 'Today' column means a winter cannot happen."

No. Open means the evidence has not established a limit or withdrawal yet — not that one is ruled out (Stage 3).

### "More users than any earlier boom means this one avoids a contraction."

No. Scale of adoption is a genuine, sourced difference, but it answers a different question than whether funding, evaluation trust, or returns hold up (Stage 4).

## Discussion questions

- If someone states a confident AI timeline today, what would you need to see before you could score it — and by when could you score it?
- The first winter's reassessment came from a commissioned report; the second came from a market collapse with no report at all. Does the absence of a report today mean nothing is wrong, or only that the evidence would look different if something were?
- Pick one "Today" claim from Stage 4. What single piece of evidence would move your confidence in it the most, in either direction?

## Questions you will get

### "So is it a bubble or not?"

Refuse the binary, in one sentence, then give the useful answer: current funding, reported usage, technical reliability, and returns are distinct questions. Earlier eras also had deployed systems and organisational users, though current consumer reach is much larger. A financial correction and a technological dead end are different claims and need different evidence.

## “Aren’t the winter dates made up?”

Partly, and the demo says so. The bands are conventions for broader contractions, not dates stamped by a single event. The 1987 hardware collapse is a useful second-winter anchor; the first has more distributed causes and withdrawals, including but not limited to ALPAC. The ends are also conventional and historians disagree; that is why the bands have soft edges and explanatory notes.

## “Where did you get this?”

Open **Settings** → **Run the self-test** in front of them. It re-derives, live, that every claim resolves to a declared source, that no source is cited and never used, and that the counts quoted on screen match the data. Then open Stage 4, which lists every source under **Every source** and, separately, everything researched and **cut** for lack of one. Both supporting sections open on request.

## “Why is that quote from a quotation-checking website?”

Because the Simon and Minsky lines are routed through a quotation-checking source that cites printed books and page references. This build has not inspected those original pages, so it labels the claims **reported** rather than primary. That distinction is part of the lesson: a plausible citation route is not the same as direct verification.

## Presenter controls

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- Turn on **Settings** → **Presentation mode** for projector-sized type.
- Run the self-test once so you know it is green before anyone asks.
- Between rooms, **Settings** → **Reset** puts the demo back to its start without reloading the page — answers cleared, Stage 1, timeline rewind, opened cards closed — and keeps Presentation mode on so the projector does not change under you.
- The app saves nothing anywhere, so an ordinary reload resets it too.
- No network is needed at any point. If the room’s wifi is down, nothing changes.

## The buttons, top right

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- ? — **Guide**: the short how-to panel that opens on first load, reopenable at any point. (Not this document.)
- ⚙ — **Settings**: **Open Presenter Notes** (this document, on screen), **Presentation mode**, **Reset**, and **Run the self-test**.

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AI Winters: Boom and Bust — Bryant Harrison, Murray State University. Sources checked September 2026. Every figure on screen links to its origin; everything that could not be sourced is listed in Stage 4 under “What was cut, and why”.